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CHE494: Entrepreneurship in Chemical Engineering

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1 Report Objectives

In preparation of a patent filing for the portable handheld bidet, this paper sought to examine and analyze the following:

1. **Patent Search** - A lookup to determine already existing and accepted patents of similar devices and what sets this current invention apart.
2. **Strategic Partners** - Companies that will help take the invention to greater heights through increased customer base, and added resources. Their respective market shares will also be discussed.
3. **Competitive Pricing Study** - To determine a price that will make the product competitive with existing products already available in the market.

2 Introduction

Portable washing devices for one's personal hygiene and/or ritual purpose are available in the market, albeit with limited choices. These devices typically come with a bottle, a delivery nozzle, as well as an optional pump. However, limitations exist with current products. Such include the overall portability or compromised hygiene. Additionally, these portable devices tend to lack certain features or components of the larger, fixed devices which reduces their overall usefulness.

Hence, a device was invented in an effort to narrow the gap between fixed bidets and portable bidets overall by incorporating certain features that allow full portability (discussed in section 3.1), better hygiene, and overall ease for the end user.

The scope of this assignment focuses on the pre-patent process, where an invention is analyzed alongside competing products and companies, potential market share, as well as the potential strategic partners that could bring the invention to a level playing field.

3 Patent Search

A patent search for similar products already filed was conducted. Since the patent office requires that all new patents filed be of an original, and new idea, these were some of the key findings that set this current invention apart from the predecessors:

3.1 Portability

Currently, most devices that claim “portability” are not really portable per se, as those products require the user to install the unit *unto* the toilet bowl itself,¹² with connections to plumbing and electricity as necessary.³ This means that the product cannot be used in outdoor or camping situations.

Furthermore, such devices are sought to be manufactured as a single unit which could mean added bulkiness. The new invention calls for the ability to disassemble the product into smaller parts which all fit nicely in a carrying case, thereby increasing convenience for the end user.

Secondly, the invention allows the user to choose any bottle he/she desires, since the cap attachment is said to be available in different configurations (plugs, screw threads, etc) that can fit an array of bottle types. All these factors combined will allow one to fully use this product in the outdoors without having any reliance on existing toilet bowls or electrical/plumbing lines.

3.2 Hygiene

A device like this seeks to increase one’s level of hygiene through thorough washing of their genitals or anuses. However, this poses a potential problem where water back-flow can contaminate the delivery tube, which therefore requires conscientious efforts in cleaning and maintenance.

All patents discovered with regards to this area have a reusable and washable delivery tube. There was even one patent found that does not require any use of tubes, rather, having an adaptor that fit directly onto the bottle which allowed water to be sprayed out directly like a watering can without a spout.⁴

Conversely, the invention incorporates *disposable* delivery tubes which alleviates the issue of cleaning and maintenance simply by disposal after using.

3.3 Power Delivery

Power is required in some way or another to deliver water out of the nozzle or tubing of the bottle - be it mechanically or electrically. Mechanically, the user is required to manually squeeze the water containing vessel to exert pressure and force the fluids out. ⁵Electrically, products are required to either be plugged into a wall socket, or batteries to function.

This product invention employs both methods, allowing the user to choose. He/she can either squeeze the bottle to manually pump the water out, or make use of an available pump as part of the kit that connects to an external, rechargeable battery power bank. Should the user choose the power bank route, the battery can be recharged and stowed away in the box.

4 Strategic Partners

Strategic partners are an important aspect in every entrepreneurial journey. They allow for small businesses to grow their customer base, access new products and markets, and block off competitors which eventually improves their business. And the same applies for said invention - Proposed strategic partnerships formed with the following companies make an ideal choice:

1. **Kohler:** Founded in 1873, Kohler Co. is one of the oldest players in the plumbing industry. With more than 10 decades of experience, they can definitely be considered as a leading brand of plumbing appliances. Currently, they do have bidets albeit available in a more traditional standalone unit - much like the ones one might find in European countries. They take up 27.4% of the current market share with respect to plumbing needs, making them the largest player in the industry as well.⁶

As such, bringing this invention and collaborating with Kohler would allow them to tap on current and new innovations, thereby expanding their repertoire of products offered while allowing the inventor of the product to tap on their experience, resources, and established customer base.

2. **American Standard:** The third largest contributor to the plumbing industry, taking up to 7.1% of the market share. While there is a significant difference in terms of market share contributions, American Standard, like Kohler, has decades of experience since their formation in the 1910s.
3. **National Defense Agencies:** While it may seem odd, this agency should be partnered with respect to targeting a specific market, mainly the military. In countries where muslim populations are of a significant presence, such as the Middle East, Malaysia, Indonesia, and Singapore, the portability of the invention would allow soldiers out in the field to be able to perform the *salat* and *wudu* rituals.

This not only enables them to conduct their religious practices, but would also significantly improve convenience and lighten their loads since they just need to be issued disposable tubes while sharing the same pump and eliminate waste accumulation at latrine points. After all, field discipline requires that no trash be left behind on jungle trails and toilet papers are extremely obvious to the naked eye if not disposed of or buried properly. Employing the said invention would completely render the need for toilet papers unnecessary, thereby maintaining the strict level of field discipline.

4. **Saniflo:** Saniflo was founded in 1958 after they started out as designers for wastewater treatment plants. A less of a common household name, a collaboration with one of the younger players in the game would allow combined and strategic partnerships to compete with the already more established companies. As of this writing, no data were available with regards to Saniflo's market share.

With the exception of National Defense Agencies, American Standard, Kohler and Saniflo were specifically chosen because plumbing forms the core of their company. Preferentially, it would be to the best of the inventor's interest if the invention could be collaborated with companies that solely focuses on providing plumbing solutions to their customers; unlike other named brands such as Moen which supplies not only plumbing collaterals, but also smart home integration solutions, and garbage disposals, etc even though they are large players of the market by taking 24.8% of the market share.

5 Competitive Pricing Study

5.1 Manual Portable Bidets

All manual products available are essentially existing for the user to squeeze the bottle so that pressure is applied to extricate water out for washing. Simple, basic, and quick. Here are the prices of some available products on the market:

** All prices listed are before tax*

1. **TUSHY Travel:** \$29 ⁷
2. **USUNO Portable Bidet Sprayer:** \$7.88 ⁸
3. **Esprite Bidet Bottle Portable Bidet Sprayer:** \$10.00 ⁹

The products sold by USUNO and Esprite are available on amazon, but TUSHY being a more upscale brand, has their product sold at close to \$30. That being said, should the customer be allowed to choose between a manual or electric operation for the bidet, the invented product can be marketed at the range of TUSHY's within a range of \$25 to \$30 to be competitive. Although the product released by TUSHY guarantees portability, the ability to fix the proposed invention to any non-comprised hygiene of the proposed invention can be a key factor that sets the competition apart.

5.2 Electric Portable Bidets

When comparing between manual and electric portable bidets, the electric market has a wider range of options ranging from functionality to even promote a lifestyle change. Here are some of the products currently available:

1. **TOTO Travel Handy Washlet:** A well established player in the toilets market since 1917, TOTO's handheld product comes with a carrying case that guarantees ease of mobility when traveling with it, and is battery operated. It even allows the user to choose the intensity of the water jet by increasing the number of holes in the outlet delivery nozzle. However, the nozzle is fixed and cannot be replaced or discarded which necessitates regular care and maintenance. \$67.99 ¹⁰

⁷<https://hellotushy.com/products/tushy-travel-bidet>

⁸https://www.amazon.com/dp/B07PNN9QJ4/ref=cm_sw_em_r_mt_dp_U_gKcXDb46GGG9X

⁹https://www.amazon.com/dp/B07XP4HL68/ref=cm_sw_em_r_mt_dp_U_dG1WDb35VZNT9

¹⁰https://www.amazon.com/dp/B005FDJ8SM/ref=cm_sw_em_r_mt_dp_U_qKVWDb1TE1780

2. **SONNY Portable Bidet:** Marketed primarily for environmentally conscious users, the product boasts long battery life (single charge lasting up to three weeks), and competitive pricing over toilet paper in the long run. It is launched in the United States as a lifestyle product, encouraging new users to switch over from traditional toilet paper to bidet. \$99.¹¹
3. **bioBidet:** The simplest form of an electronic bidet. A delivery nozzle folds out of the body and the user presses a button to initiate the flow of water. \$37.05¹²

Given that the proposed invention comes with a pump and a power bank in the kit, coupled with the added advantages that this proposed invention carries over its competitors, a reasonable price to sell this product with the power bank would be in the range of \$88-\$99.

By starting at \$88, it makes for a reasonable additional top up from the TOTO product should the idea of disposable delivery tubes be of a huge interest to the customer. If the product is priced at \$99, it brings itself to the same level as that of the product released by SONNY. The attraction towards said invention can stem off a more comprehensive portable bidet system that offers more features for the buck. However, it is not recommended to go below \$80 as added costs for the pump and battery pack would therefore lower the overall profit.

¹¹<https://www.indiegogo.com/projects/sonny-a-portable-bidet-for-eco-conscious-humans/>

¹²https://www.homedepot.com/p/bioBidet-Electric-Portable-Travel-Bidet-in-White-TP-200/300358803?mtc=Shopping-B-F_D29B-G-D29B-29_32_TOILETS_AND_SINKS-Multi-NA-NA-PLA-NA-NA-Toilet_Seats&cm_mmc=Shopping-B-F_D29B-G-D29B-29_32_TOILETS_AND_SINKS-Multi-NA-NA-PLA-NA-NA-Toilet_Seats-71700000040795031-58700004374350564-92700048656978135&gclid=Cj0KCQiA-4nuBRcNARIsAHwyuPqbuWFQgbwaoTpgV8MeQ-WqI4tovkV0i7wiLXYHU59NksEOaA4Ym2gaAqx-EALw_wcB&gclidsrc=aw.ds

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